Time-Aligned Lyrics for Ethiopian Music

A Dataset Design and Annotation Workflow for Amharic Songs

Yonatan Assefa Mekonnen
Independent researcher, Addis Ababa, Ethiopia
yonatanassefa60@gmail.com
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Document status. Preprint, version 1.0, 14 October 2026. Not peer reviewed. This paper proposes a dataset design and an annotation workflow. It reports no collected corpus, no annotator study, and no experimental results. The prototype is a proof of concept, not a production annotation service.

Abstract

A written lyric shows which words are sung but not when each word begins and ends in a recording. Word-level timestamps support synchronised lyrics, lyric search, music education, and research on sung language. Word-level lyrics-to-audio alignment is an established research task with public benchmarks in European languages, and Ethiopian vocal music already has a word-timed resource in liturgical Ge'ez chant (the Zema Dataset). This paper therefore makes a narrower claim. In the sources reviewed, a public and documented collection of contemporary Amharic songs with verified word-level timings was not found, and the conventions, reliability, effort, and rights handling needed to build one have not been worked out. The paper proposes a feasibility-oriented design for such a resource: a draft record schema; Amharic-specific tokenization and annotation conventions for melisma, repetition, overlapping voices, and code-switching; a ten-step rights-aware annotation workflow; a minimal architecture; a client-side proof-of-concept annotation interface; contributor and quality-control mechanisms; and an evaluation protocol with pre-declared timing metrics and tolerances. The review of related work is targeted, not systematic, and the availability of the Zema data and the performance of multilingual aligners on Amharic singing remain to be established. Only the schema draft and the prototype exist, and no empirical results are reported. A small rights-cleared pilot is identified as the next step.

Keywords: Amharic; Ethiopian music; lyrics-to-audio alignment; singing; music information retrieval; dataset design; annotation; low-resource languages

1. Introduction

A written lyric and a music recording carry related but different information. The text says which words are sung. The recording says when each word begins and ends, how long it is held, how it is pronounced, and how it sits against the accompaniment. A listener who knows the song can usually follow the words, but turning that knowledge into reliable word-level timestamps requires repeated listening, boundary editing, and a consistent policy for awkward cases. At any scale this work is slow and inconsistent unless the task is supported by a clear interface, written conventions, and quality control.

Word-level lyrics-to-audio alignment is an established research task. Given a recording and its lyric transcript, a system assigns a start and end time to each word. The MIREX evaluation campaign has run the task for several years and, in its 2026 description, specifies word onset and offset times as the required output [1]. Methods range from speech-recognition models adapted to singing [2] to neural systems that jointly separate the voice and align phonemes [3], predict characters directly from polyphonic audio [4], or learn audio and text embeddings with a contrastive objective [5]. These methods are evaluated on a small number of public benchmarks, and the benchmark languages are overwhelmingly European [1, 7]. Whether they transfer to Amharic singing is not something the literature I reviewed can answer.

This paper starts from a practical observation by the author, a software engineer in Addis Ababa: building a reusable word-timed lyrics resource for contemporary Ethiopian songs is, at present, an open design problem rather than a solved one. Two resources must be acknowledged at the outset because they bound what can honestly be claimed. The Zema Dataset documents roughly ten hours of Ethiopian Orthodox Tewahedo Church chant, with word-level temporal boundaries, in the liturgical language Ge'ez [10]. Dataset.ET has released 22.7 hours of Amharic read speech under a CC BY 4.0 licence [12]. Neither is a collection of contemporary Amharic songs with word timings, and neither claims to be. They also show that the surrounding infrastructure is beginning to exist, so the argument here is not that Ethiopian language technology is unexplored.

The paper is a **research design and technical report**. It does not report a collected corpus, an annotator study, a benchmark, or any measured result. It contributes a worked-out specification (schema, conventions, workflow, evaluation protocol, rights policy) and a small browser prototype of the annotation interaction, and it states plainly which parts exist and which are proposals. Section 1.2 gives the status of each contribution.

1.1 Research questions

The paper addresses three questions, chosen so that each can be answered, at least in part, by design work and literature review alone, and each can be tested later by a pilot.

- **RQ1.** What metadata and annotation structure does a reusable word-level lyric-alignment resource for Amharic songs need, and how should Amharic orthography and singing phenomena (melisma, repetition, overlapping voices, code-switching) be represented?
- **RQ2.** What annotation workflow and tooling can a single developer realistically build and a small group of fluent contributors operate, while preserving provenance, review, and rights information?
- **RQ3.** What evaluation protocol would establish whether the resulting annotations are reliable and affordable, and what would be needed to decide between manual, automatic, and hybrid annotation?

Rights and ethics are treated as design constraints on all three questions rather than as a separate afterthought, because they decide whether a dataset can be released at all.

1.2 Contributions and their status

Table 1 lists each contribution together with its status. Only the prototype and the draft schema exist as artefacts. Everything else is a proposal to be tested.

Table 1. Contributions and their current status.

Contribution	Description	Status
Resource landscape review	Targeted review of lyrics alignment, singing transcription, and Ethiopian language and music resources (Sections 2 and 3).	Done, but not systematic. Search log in Appendix D.

Contribution	Description	Status
Schema v0.1	Record and word-segment fields, controlled values, validation rules (Section 5).	Drafted. Partly enforced by the prototype.
Annotation conventions	Rules for tokenization, boundaries, melisma, repetition, overlap, code-switching (Section 5, Appendix B).	Proposed. Not yet tested with annotators.
Workflow and architecture	Ten-step rights-aware workflow and a minimal architecture (Sections 6 and 7).	Proposed.
Prototype v0.1	Client-side React and TypeScript tool: local audio, waveform, word timing, tap-along mode, cosmetic game layer, JSON export (Section 7).	Implemented. Build tested. One scripted browser test. No user study.
Evaluation protocol	Metrics, tolerances, sampling, agreement and cost measures (Section 9).	Proposed. No results exist.
Rights and ethics policy	Collection, consent, release, and takedown principles (Section 10).	Proposed. Legal questions listed for local counsel.
Pilot, dataset, model comparison	Rights-cleared pilot, annotation study, comparison with automatic aligners.	Future work. Not started.

2. Background and related work

This section reviews the literatures that bear on the design and says, for each, what the design takes from it. The review is targeted rather than systematic. How the searches were run, and what remains unchecked, is recorded in Appendix D.

2.1 Lyrics-to-audio alignment

Lyrics-to-audio alignment estimates where supplied words occur in a recording. It differs from automatic lyrics transcription, in which a system must infer the words without a transcript. The two are related, but a transcript can contain the right words with wrong timing, and precise timestamps can be attached to the wrong words. The proposed schema therefore keeps text status and timing status separate.

The MIREX 2026 task description defines the input as mixed singing audio with word-level lyrics and the output as word onset and offset times in seconds [1]. Its listed collections are the Jamendo test set (20 pieces, held out), Hansen's set (9 songs with word begin and end times, mixed and a cappella), and Mauch's set (20 songs with word onsets), with DAMP, DALI, and Gracenote named as optional collections. The page names English for the Hansen and Mauch songs and does not state the languages of the others. It ranks systems by average absolute timing error, by the percentage of correctly aligned duration, and by the percentage of estimates within a 0.3 s tolerance [1]. These are the metrics that Section 9 adopts, with additions, so that results on a future Ethiopian set can be read against the existing literature.

Early work adapted speech recognisers to singing. Mesaros and Virtanen trained a phonetic hidden Markov model on speech, adapted it to singing, used vocal separation for polyphonic audio, and reported an average line-level alignment error of 0.94 s [2]. Gupta et al. used a speech recogniser and published lyrics to align lyrics to solo singing semi-automatically, and suggested that the approach could help build a large aligned corpus [9]. Later systems are end-to-end. Stoller et al. predict characters from raw polyphonic audio and train with weak, line-level annotations [4]. Schulze-Forster et al. combine phoneme-level alignment with text-informed voice separation [3]. Durand et al. use contrastive learning, report an average absolute error below 0.2 s on the standard Jamendo set, and claim robustness to languages not seen in training [5]. These claims are the authors' own, on their own benchmark, and are not evidence about Amharic.

What the design takes from this literature. Alignment quality depends on the language model, the pronunciation resources, the acoustic domain, and the presence of accompaniment. Public implementations are typically tied to English tooling. The released lyrics-aligner repository, for instance, accepts phonemes from the 39-symbol ARPAbet set and converts words through the CMU dictionary, and its authors expect errors in

challenging mixtures [18]. Automatic output should therefore be treated as a hypothesis to be verified by a person, and its usefulness for Amharic should be measured, not assumed.

2.2 Aligned-lyrics datasets

JamendoLyrics provides word-level and line-level start and end times and is described by its maintainers as primarily an alignment benchmark [7]. The multilingual version contains 79 songs: 20 English, 20 German, 20 Spanish, and 19 French. Each song carries its own licence type, with values such as CC BY-NC-ND, CC BY-NC-SA, and CC BY-SA visible on the dataset page [7]. The original English-only set was introduced alongside the end-to-end alignment work of Stoller et al. [4], and the multilingual extension accompanies Durand et al. [5]. Cífka et al. later revised the same material into the Jam-ALT benchmark to follow industry lyric-formatting conventions, which shows that the conventions for what counts as the lyric text are themselves a design choice [6].

DALI is far larger. Its second version holds 7,756 songs and 488.1 hours of audio, covers 32 languages, and provides time-aligned lyrics at note, word, line, and paragraph level together with vocal melody notes [8]. Its audio is retrieved from YouTube rather than redistributed, which is one precedent for releasing annotations without audio. Whether that approach is acceptable for a given rights holder or jurisdiction is a legal question, and the DALI paper does not settle it for this project. I did not establish whether DALI contains any Amharic songs.

What the design takes from this literature. Per-song licence metadata (as in JamendoLyrics) and annotation-only distribution (as in DALI) are both live models. The proposed schema records the licence and permitted uses for every recording so that a release can be assembled from whatever subset the rights allow.

2.3 Ethiopian language and vocal-music resources

Speech resources for Ethiopian languages predate recent efforts. Abate et al. built read-speech corpora of about 22 hours for each of four languages (the abstract's wording is ambiguous as to whether this is per language or in total) and trained baseline recognisers, reporting a word error rate of 37.65% for Amharic [13]. Dataset.ET has released 22.7 hours of Amharic read speech, 7,405 recordings from 320 speakers, under CC BY 4.0 on Hugging Face, managed by Snapwre Technologies PLC in Addis Ababa [12]. Gete et al. fine-tuned Whisper for Amharic using FLEURS, Common Voice, and a further corpus, and found that mixing existing and new data mattered and that homophone normalisation improved word error rate [14]. The Dataset.ET source is a project blog post and not a peer-reviewed publication, and the figures are the project's own.

These resources are read speech. Singing changes duration and pitch, can obscure consonants, and often occurs over accompaniment, so a speech corpus cannot be assumed to represent the conditions of a song. They are nonetheless relevant in two ways: they are candidate training or adaptation data for Amharic acoustic models, and they show that community-contributed collection of Ethiopian language data is feasible at the scale of tens of hours.

The Zema Dataset is the closest prior work and is discussed in detail in Section 3. Massively multilingual speech models are also relevant. Pratap et al. report a CTC-based alignment model that supports 1,130 languages, built to segment long recordings [16]. Whether Amharic is among those languages, and how such a model behaves on singing, are questions this paper does not answer, but they make the model a natural candidate for the automatic-suggestion step evaluated in Section 9. The Montreal Forced Aligner is a widely used speech aligner [17]. Its availability for Amharic and its behaviour on singing were not checked.

2.4 Amharic orthography and its consequences for annotation

Amharic is written in the Ethiopic script, in which each basic character represents a consonant with a vowel. Several consonants have more than one character with the same pronunciation, such as ሀ, ሐ, ኀ and ኸ, and the same holds for አ and ዐ, ሰ and ሠ, and ጸ and ፀ. Belay et al. document that writers use these homophones inconsistently, that normalising them can change meaning in some words, and that the effect of normalisation depends on the task: in their experiments it hurt part-of-speech tagging and sentiment analysis and helped retrieval [15]. For this project the practical consequence is that a lyric transcript can be correct and still vary in spelling between sources. The schema therefore stores the verified original text and an optional normalised form, never the normalised form alone.

Whitespace splitting is a reasonable first tokenizer for Amharic text, but it yields orthographic words, not morphemes. Common prepositions and conjunctions are usually written attached to the following word, and pronominal suffixes attach to verbs and nouns. A fluent reviewer should confirm that the orthographic word is the right annotation unit for the intended uses (this is a linguistic judgement to be checked, not a result established here). Section 5.3 specifies the unit and the escape hatches.

2.5 Annotation practice, crowdsourcing, and dataset documentation

Two bodies of work inform the contributor model and the documentation. Fort et al. review paid crowdsourcing on Amazon Mechanical Turk for language resources, report observed wages below US$2 per hour, argue that higher pay does not reliably improve quality because weak reputation systems let spammers through, and call for fair-wage platforms and ethics review of crowdsourced work [21]. That is evidence about one platform and not about timing annotation, and it validates no particular incentive for this task. For documentation, datasheets for datasets [19] and data statements for language data [20] provide checklists that the eventual release should follow, and Section 10 maps them onto the proposed metadata.

3. Resource landscape and gap analysis

The claim I am willing to defend must be as narrow as the evidence. Table 2 summarises what the sources I read establish about each resource, and then the gap statement follows.

Table 2. Resources most relevant to the proposal, as described in their own documentation. The last column states what the resource does not establish for this project.

Resource	Content and scale	Language and domain	What it does not establish here
Zema Dataset [10, 11]	About 10 h, 369 instances. Word-level temporal boundaries, reading-tone and chanting-mode labels. Annotated in Audacity, recorded in spreadsheets.	Ge'ez liturgical chant. Audio from one book (Se'atat Zema), by one scholar, from an audiobook site.	Contemporary songs. Amharic lyrics. Inter-annotator agreement (none reported). Licence (none stated). Whether the public repository contains the full audio and all word timings (see text).
JamendoLyrics [4, 5, 7]	79 songs (multilingual). Word and line timings. Per-song licences.	English, German, Spanish, French. Mostly Jamendo-hosted music.	Any Ethiopian language.
DALI v2 [8]	7,756 songs, 488.1 h. Note, word, line, paragraph alignment.	32 languages, mostly English pop and rock. Audio retrieved from YouTube.	Amharic coverage (not checked). Rights position for redistribution.
MIREX test sets [1]	Jamendo (20), Hansen (9), Mauch (20). Optional: DAMP, DALI, Gracenote.	English for Hansen and Mauch. Others not stated on the page.	Any Ethiopian language.
Dataset.ET [12]	22.7 h, 7,405 recordings, 320 speakers. CC BY 4.0.	Amharic read speech.	Singing. Word timings. Music.
Abate et al. corpora [13]	About 22 h (per language or total, unclear). Baseline ASR.	Amharic, Tigrigna, Oromo, Wolaytta read speech.	Singing. Music.

What the Zema Dataset shows. Its paper reports that annotators opened each recording in Audacity, zoomed, listened, and typed word start and end times into a spreadsheet; that three annotators labelled about six hours; that a certified Ge'ez expert annotated reading tones for all 10.21 hours; and that annotators cross-reviewed each other and inspected mel-spectrograms with word labels overlaid. The annotators reported about 200 hours for first-round annotation and about 100 hours for double-checking. No inter-annotator agreement statistic is reported, and the paper's own preliminary mode-classification experiment is described as a first step [10]. These details are valuable twice over: they show that word-level timing of Ethiopian vocal music has been done by a manual process, and they give a rough effort benchmark (Section 8.4).

The public repository named in the paper contains a CSV of mode labels and a small supplementary folder with three sample recordings, and no licence file that I could find [11]. Because that listing may not reflect the full release, I did not establish whether the full audio and word-level timings are publicly downloadable, or under what terms. This should be checked directly, ideally by contacting the Zema authors, before any statement about its availability is published.

3.1 Gap statement

The evidence supports the following statements, and no stronger ones.

1. Word-level timed vocal data for Ethiopian music exists in at least one domain, liturgical Ge'ez chant [10].
2. In the sources I searched, I did not find a public, reusable collection of contemporary Amharic songs with verified word-level timestamps, nor a documented annotation convention for sung Amharic. This is a statement about a non-exhaustive search conducted in October 2026, not proof of absence.
3. Public lyrics-alignment benchmarks and the models evaluated on them cover European languages, so their behaviour on Amharic singing is untested in the literature I reviewed [1, 4, 5, 7].
4. Where a relevant resource exists, its coverage, licence, annotation reliability, or availability may not be established (the Zema case). Those are separate dimensions on which a new resource could differ, and each should be verified before being claimed.

The contribution proposed here is therefore not a claim of novelty at the level of the task. It is a feasibility-oriented design for a different musical domain (contemporary Amharic songs), with explicit conventions, reliability measurement, and rights handling, to be tested by a pilot. If a later, more systematic search finds an existing resource, the design should be repositioned as complementary.

4. Problem definition and scope

The task is: given a recording and a verified lyric transcript of what is sung, assign each transcript word a start and end time in seconds from the beginning of the audio file. The start marks the earliest audible evidence of the word, and the end marks the end of its audible realisation, including any sustained vowel, before the next word or silence. These definitions are operational and are expanded in Appendix B.

The transcript is a transcript of the *performance*. Published lyric sheets often differ from what is sung, because singers repeat lines, improvise, elide syllables, or substitute words. If the verified transcript does not match the audio, timing cannot be assigned. The workflow therefore includes a transcript-checking stage before timing begins (Section 6), and tokens are frozen before annotation so that independent annotators label identical units.

The first phase is limited to Amharic-language songs for which transcript quality can be checked and rights can be documented. Other Ethiopian languages, liturgical repertoires, and traditional forms differ in orthography, performance practice, and cultural context. Extending to them requires fluent reviewers and language-specific conventions, and in the case of religious material, consultation with the relevant communities. The paper does not claim that the Amharic conventions transfer.

Three tasks are explicitly out of scope: transcribing lyrics from audio, training or releasing models, and annotating pitch, rhythm, or musical structure. Phoneme-level boundaries, as in Schulze-Forster et al. [3], are also out of scope for the first version, because they raise annotation cost and need a pronunciation convention for Amharic that has not been established.

5. Dataset schema and annotation conventions

This section answers RQ1. The schema is deliberately small. Fields were added only where a stated use, a quality-control need, or a rights obligation requires them, following the principle (also present in the Zema paper's choice to add labels only where it had a use for them [10]) that unclear labels cost more than they yield.

5.1 Record structure

A record describes one recording and the lyric segments timed against it. It has five groups of fields, listed in Table 3. Timestamps are numeric seconds from the first sample of the audio file as distributed, rounded to millisecond precision. Audio is identified by checksum so that timings cannot silently be applied to a different encoding of the same song.

Table 3. Proposed schema v0.1. Fields marked R are required. The prototype currently exports the subset shown in bold.

Field	Content	Notes
recording_id (R)	Stable identifier.	Opaque. No personal data.
audio.file_name, **audio.duration_s**, **audio.sha256** (R)	File name, duration, SHA-256 of the distributed file.	Also sample rate and format. Checksum guards against re-encoding mismatches.
audio.source_type	One of: commissioned, artist_licensed, open_licence, label_licensed, other_documented.	Determines what may be released (Section 10).
rights.* (R)	Rights holder(s) for the recording and, separately, for the lyrics. Permitted uses. Redistribution status. Licence. Evidence reference.	Recording and lyric rights are recorded separately. Uses: annotate, store, release annotations, release audio, train models, commercial use.
language, script (R)	BCP-47 tag for the recording (for example am) and script code (Ethi).	Segment-level lang for code-switching.
lyrics.text_original (R)	Verified transcript of what is sung, in original spelling.	Never overwritten. Corrections create a new version.
lyrics.text_verified_by, lyrics.tokenization_version	Pseudonymous reviewer ID. Tokenizer and rule version.	Tokens are frozen before timing starts.
lyrics.normalization	Whether a normalised form was produced and which mapping was used.	Optional. Homophone mapping, if any, is documented, not implied [15].
segments[].index, text (R)	Position and surface form of the token.	Repeated words are distinct segments.
segments[].start, end (R)	Seconds. Required: 0 <= start < end <= duration.	Overlap with other segments is allowed only across layers.
segments[].text_normalized	Optional normalised token.	Null when not normalised.
segments[].lang	Language tag per token.	For code-switching. Default is the recording language.
segments[].layer	lead, backing, or unknown.	Backing vocals are annotated only when a policy and a reviewer exist.
segments[].flags	Controlled values: melisma, repeated_ornament, elided, spoken, uncertain_text, uncertain_time, overlap.	Free-text note allowed. Flags drive review sampling.
segments[].status	untimed, timed, reviewed, uncertain.	Prototype values. A production system would also track accepted and rejected.
segments[].confidence	Optional annotator confidence (low, medium, high).	Self-report only. Not a measure of accuracy.
annotation., *review*.	Pseudonymous annotator and reviewer IDs, tool version, timestamps, revision history, outcome.	Payment details are stored elsewhere, never in the record.

A synthetic example record is shown in Figure 1. It was written by hand to illustrate the structure. It does not come from any recording, the timestamps are placeholders, the audio file does not exist, and the record is marked `is_sample: true`. A matching file is included in the prototype repository.

```
{
  "schema_version": "0.1",
  "is_sample": true,
  "recording": {
    "recording_id": "SAMPLE-0001",
    "audio": { "file_name": "synthetic-demo.wav", "duration_s": 4.0,
               "sha256": null, "sample_rate_hz": 16000 },
    "language": "am", "script": "Ethi",
    "rights": { "status": "illustrative_placeholder", "rights_holder": null,
                "permitted_uses": [], "redistribution": "none" }
  },
  "lyrics": {
    "text_original": "ሰላም ኢትዮጵያ",
    "text_verified_by": "placeholder",
    "tokenization_version": "ws-1",
    "normalization": { "applied": false }
  },
  "segments": [
    { "id": "s0", "index": 0, "text": "ሰላም", "text_normalized": null,
      "start": 0.42, "end": 0.91, "lang": "am", "layer": "lead",
      "flags": [], "confidence": null, "status": "timed" },
    { "id": "s1", "index": 1, "text": "ኢትዮጵያ", "text_normalized": null,
      "start": 1.04, "end": 1.88, "lang": "am", "layer": "lead",
      "flags": ["melisma_possible"], "confidence": null, "status": "timed" }
  ],
  "annotation": { "annotator_id": "pseud-A01", "tool": "ethio-lyric-align 0.1",
                  "created_utc": null, "revisions": [] },
  "review": { "reviewer_id": null, "outcome": null, "notes": null }
}
```

Figure 1. Synthetic schema example (not a real annotation).

Validation rules follow from the fields. Structural checks are: start and end present together; start >= 0; end > start; end <= duration; no segment longer than a configurable ceiling (the prototype flags intervals over 10 s for review); token count equal to the frozen token list; and start times non-decreasing within a layer. Overlap across layers must not be rejected, because backing vocals and call-and-response legitimately overlap the lead. These checks catch structural errors only and cannot show that a boundary matches the audio.

5.2 Versioning and provenance

Every release carries a version number, date, change log, licence, and citation text. Annotation edits are stored as revisions, not overwrites, so that disagreement and correction history can be analysed and so that a contributor's original work can be audited. If permission for a recording is withdrawn, the record is flagged and removed from later releases, and the change log states that a removal occurred without exposing private details.

5.3 The annotation unit and Amharic tokenization

The annotation unit is the orthographic word: a maximal run of non-space characters in the verified transcript, after removing sentence punctuation. Ethiopic punctuation (the word separator ፡, full stop ።, comma ፣, semicolon ፤, colon-like marks ፥ and ፦, and question mark ፧) is stripped from token edges and kept in the text field, never as a separate interval. The prototype implements this rule with whitespace splitting and a character-class filter, and a contributor can correct a bad split by editing the lyric text and re-tokenizing. Four further decisions are recorded.

- **Orthographic, not morphological.** Attached prepositions, conjunctions, and suffixes remain part of the word they are written with. This keeps the unit identical to what a reader sees. A morpheme-level layer would be a later addition if a user need appears.

- **Spelling preserved.** The verified transcript keeps the spelling the lyric source used. Homophone variants are not merged in the primary text [15]. If a normalised form is useful, it is added as a separate field with a documented mapping.
- **Colloquial and dialectal forms preserved.** Contributors do not convert sung forms into formal written forms unless a normalisation policy for that step has been published.
- **Tokens frozen before timing.** Once a reviewer has approved the token list, annotators cannot add, delete, or split tokens. They can flag a token as `uncertain_text`, which routes the excerpt back to transcript review. This keeps independent annotations comparable token by token.

5.4 Difficult cases

Table 4 states the proposed rule for each difficult case and the field that records it. The rules are first drafts. The pilot is intended to reveal where they are ambiguous, and Appendix B reproduces them in the form an annotator would see.

Table 4. Proposed conventions for difficult cases (draft, untested).

Case	Proposed rule	Recorded as
Sustained vowel or melisma	The word interval runs from the first audible onset of the word to the end of the last audible note or sustained vowel that belongs to it, even when it spans several pitches.	end time. Flag melisma.
Repeated word or line	Each sung occurrence is a separate segment, in transcript order. If the transcript lists a repeat once but the singer repeats it, the transcript is corrected first.	Separate segments.
Repeated syllable as ornament	If it is plainly part of the word, include it in the interval. If it is a separable vocalisation without lexical content, leave it untokenized and note it.	Flag repeated_ornament.
Overlapping voices, backing vocals	Pilot version: annotate the lead vocal only. Backing words are marked in the transcript as excluded. Later versions may add a layer field.	layer; excluded token note.
Code-switching	Tag the language of each token. If unsure, mark unknown.	segments[].lang.
Instrumental break	No segments are created for instrumental sections. A gap between segments is valid.	None.
Elision and run-together words	If two written words are sung as one acoustic unit, assign the same interval to each only if they cannot be separated, and flag.	Flag elided.
Spoken or shouted words	Time them as words and flag them, because acoustic models trained on singing may treat them differently.	Flag spoken.
Uncertain lyric or boundary	Never guess. Mark uncertain and add a note. Reviewers resolve.	Flags uncertain_text, uncertain_time.
Several defensible boundaries	Choose the earliest clear audible evidence for start and the last clear audible evidence for end, and note the ambiguity.	Flag uncertain_time. Used in disagreement analysis.

Some of these cases produce irreducible disagreement. A consonant at the end of one word can be masked by the onset of the next, and two careful annotators can place the boundary tens of milliseconds apart for defensible reasons. That is one reason the evaluation plan reports tolerance-based agreement and not only exact matches (Section 9).

6. Annotation methodology

This section and the next answer RQ2. Figure 2 shows the proposed workflow, which has ten steps. It is a design. Only part of step 4 and part of step 8 are covered by the prototype.

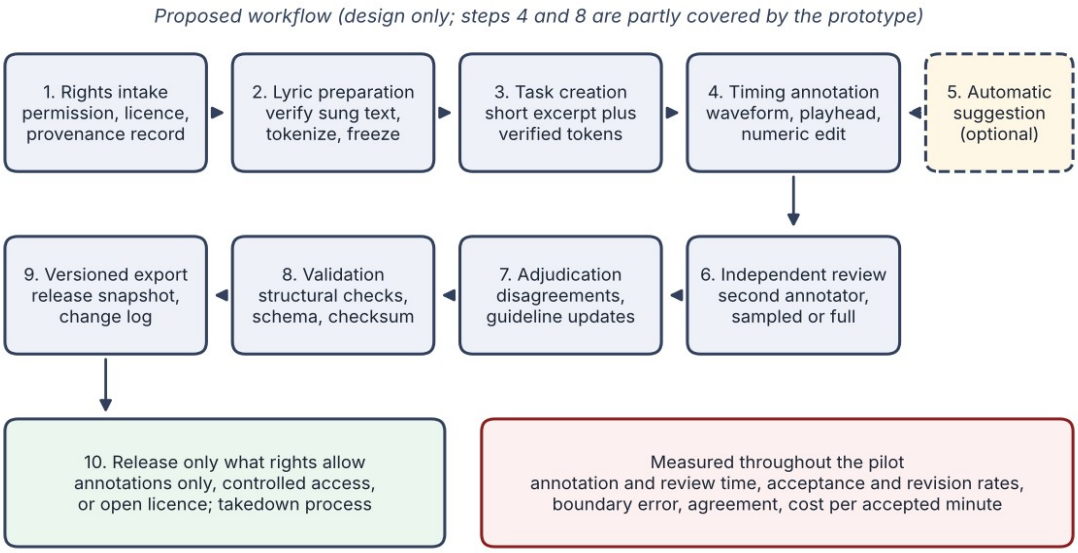


Figure 2. Proposed rights-aware annotation workflow (design only).

6.1 Sources of audio and lyrics

Rights intake comes first because it limits every later step. Table 5 compares the options in terms of rights clarity, effort, and the data the project would be able to release. Option A, commissioned recordings, gives the cleanest rights and a controllable mix of difficulty, at the cost of recording expense and limited stylistic reach. Option E is excluded: publicly accessible audio does not carry permission to copy it, redistribute it, publish its lyrics, or use it to train models (Section 10).

Table 5. Candidate sources of recordings and lyrics.

Source	Description	Rights clarity	Release possibilities
A. Commissioned	Singers record original or consented songs for the project under a written agreement.	High, if the agreement covers lyrics, recording, and uses.	Annotations and audio, as agreed.
B. Artist-licensed	Artists or their representatives grant a documented licence for annotation and a defined release.	High, per recording.	As licensed.
C. Open-licence	Music already released under an open licence, with the licence terms checked per song.	Medium to high. Terms vary and may exclude derivatives or commercial use [7].	Depends on licence.
D. Label or collecting-society agreement	A negotiated agreement for research use.	High but slow to obtain.	As negotiated.
E. Publicly available without a licence	Streaming or video-platform audio with no permission.	Not established.	Excluded from the design.

For a first pilot of a few dozen minutes, option A or B is realistic for an individual. It also removes dependence on unverified lyric sources, because the contributor or artist can supply and confirm the sung text.

6.2 Transcript preparation and task creation

The transcript is checked by a fluent speaker against the audio, because the sung text, not the published text, is what is aligned. Token frozen status is recorded. Songs are cut into excerpts of roughly 30 to 90 seconds at natural phrase boundaries. Long tasks cause fatigue and make review expensive. Where a song is split, excerpts overlap by one phrase so that boundaries can be reconciled. The exact excerpt length is a parameter to be tuned by the pilot, not a finding.

6.3 Timing and review

The annotator loads the excerpt, sees the verified tokens, and sets start and end times by listening and using the waveform. A waveform shows energy and pauses but cannot identify words, so playback and replay of short regions remain the primary evidence. A second annotator independently times the same excerpt without seeing the first result. In the pilot, which is small, every excerpt can be double-annotated, which makes agreement measurable on all of the data and gives a reference for error analysis. A reviewer resolves disagreements above a predefined threshold and records which convention applied. Review comments feed back into the guidelines, which are versioned.

6.4 Manual, automatic, and hybrid annotation

Three configurations should be compared in the pilot. In the manual configuration, annotators time every word from scratch. In the automatic configuration, an aligner proposes times for all words and a reviewer accepts or corrects them. In the hybrid configuration, an aligner proposes times only for words where an agreement or confidence check passes, and annotators time the rest. Candidates include the CTC-based multilingual alignment model of Pratap et al. [16] (if Amharic is covered), the Montreal Forced Aligner [17] (if an Amharic acoustic model and dictionary can be built or found), and English-trained singing aligners [3, 4, 5, 18] as a transfer baseline, optionally preceded by vocal separation. English-centred tooling [18] suggests that transfer will need checking, not assuming. The pilot records the model version, preprocessing, compute environment, failure cases, and the human correction time per audio minute. Automatic output is only useful if correcting it is faster than annotating from scratch, and that comparison needs the cost data of Section 8.

7. Prototype and architecture

7.1 What exists

A small prototype, ethio-lyric-align, demonstrates the interaction for step 4. It is written in React and TypeScript with Vite and runs entirely in the browser, so it can be hosted as static files. A contributor selects a local audio file, pastes lyrics, and splits them into word tokens. The audio is decoded locally with the Web Audio API, peak values are computed, and a canvas waveform is drawn with the timed words shaded. The contributor can play, pause, seek, change playback speed, select a word, set its start or end to the current playhead or type a value, play the selected word, mark it reviewed or uncertain, add a note, and export all segments as JSON. A tap-along mode lets the contributor play the song and tap as each word begins to produce a rough first pass, and a word can be marked checked only after it has been replayed. An optional game layer shows experience points, levels, and badges for timing and checking words. It is stored only in the browser's local storage, rewards process steps and not speed, is cosmetic (it is not a payment or an accuracy score), and has not been tested as an incentive (Section 8.2). Structural checks from Section 5.1 are shown beneath the word list. Figure 3 shows the interface.

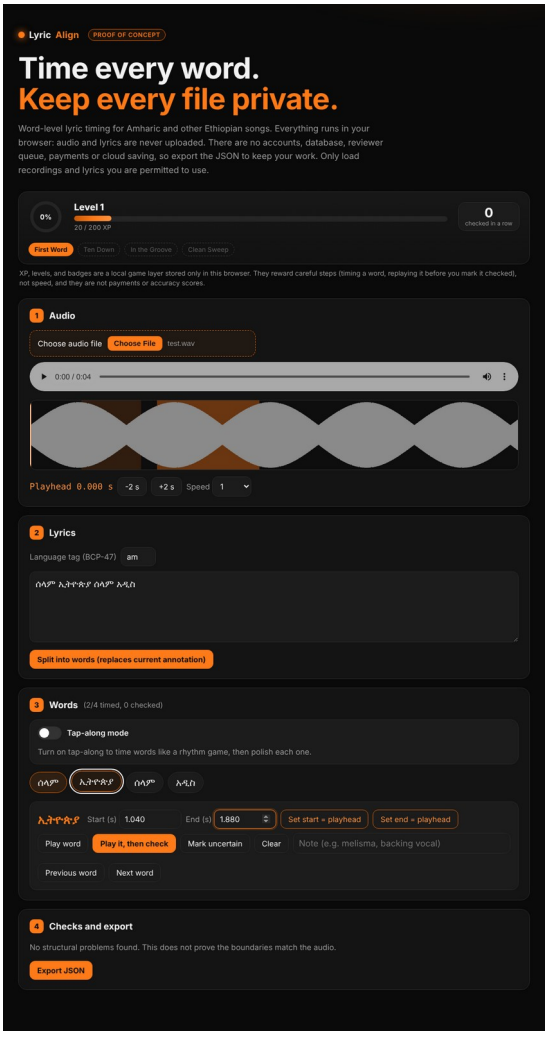


Figure 3. The prototype interface (dark theme). The audio is a generated test tone, the lyric is a short placeholder phrase, and the timestamps were entered to test the tool. The experience-point panel at the top belongs to the optional game layer.

A demonstration build is hosted at https://time-aligned-lyrics-for-ethiopian-m.vercel.app/. The prototype does not upload audio or lyrics, has no accounts, database, reviewer queue, payment function, rights check, or cloud persistence, and loads the full decoded file in memory, so it is only suitable for short excerpts. Verification so far is limited: the project builds, and scripted headless-browser runs loaded a generated audio file, tokenized Amharic text, and set word times by typing. It has not been tested by real annotators, on real songs, or on mobile devices, and no usability measurement exists. It should therefore be described as a proof of concept.

7.2 A minimal production architecture

If the pilot supports building a collection service, the following would be added in stages, each only when the pilot shows it is needed. First, local save and resume in the browser, and a task manifest file listing excerpts, tokens, and rights metadata. Second, server persistence: a relational database (PostgreSQL is a reasonable choice) for tasks, annotations as versioned rows, rights records, and review outcomes, plus object storage with access control for audio. Third, background jobs to produce waveform peaks and run permitted automatic aligners, a review queue, and an audit log. Fourth, payment records kept separately from annotation data. Each addition raises security and privacy obligations, in particular consent records and access control for contributor data and for audio that the project is licensed to hold but not to release. None of these is built. They are listed so that the schema and workflow decisions made now do not preclude them.

The prototype's design choices follow from this ordering: static hosting, local files, and no server-side storage by default, which also means that no rights-sensitive audio leaves the contributor's device.

8. Contributors, incentives, and operations

8.1 Who annotates

The task needs fluent Amharic speakers who can understand the lyric and can use a simple interface. It does not need musical training, but it benefits from careful listening. The pilot should deliberately include contributors with different backgrounds so that dependence on annotator experience can be measured. Annotation and review are separate roles, and no one reviews their own work.

8.2 Possible incentives

Possible mechanisms are transparent payment per accepted task, mobile-money or credit-based compensation, short calibration tasks with feedback, quality-based contributor levels, progress displays, and non-monetary recognition. None has been validated for this task. Fort et al. report that higher pay does not reliably improve quality in paid crowd work on one platform [21], which is a reason to base rewards on accepted work and not on volume, and to be cautious about leaderboards or streak mechanics that favour speed. They also raise concerns about low pay and the lack of worker protections [21], which argue for stating the rate, the acceptance criteria, the review timeline, and the appeal process before work starts. The prototype's optional game layer (Section 7.1) is a cosmetic version of the progress and badge ideas and does not validate them.

8.3 Quality control and abuse

Risks include copied timestamps, careless submission, fabricated lyrics, account farming, collusion, and reviewer fatigue. Mitigations are calibration tasks, planned duplicate annotation, random and targeted audits, timestamp sanity checks, rate limits, and a clear path for contributors to contest a rejection. Automatic flags should prompt review, not accusations. Contributor data should be minimal. Public releases should use pseudonymous identifiers unless contributors ask for attribution, and payment details should be stored apart from annotation records.

8.4 Cost, sustainability, and an effort benchmark

The principal cost measure is cost per accepted audio minute, which accounts for rejected work and review. Its components are annotation time, review time, rework, rights clearance, tooling, and coordination. As an arithmetic template:

```
cost per accepted minute = (annotator hours x annotator rate + reviewer hours
          x reviewer rate + fixed costs) / accepted audio minutes
```

The only external effort figure I found is from the Zema paper, whose annotators reported about 200 hours for a first round and about 100 hours for double-checking on a dataset of about 10.21 hours [10]. The paper does not say precisely which portion of the data each figure covers (three annotators timed about six hours, and a separate expert annotated reading tones for all of it), so any ratio is rough. If the full 300 hours is taken as the effort for the full 10.21 hours, it is about 29 hours per audio hour. That figure belongs to a different repertoire, a different task bundle, and an unknown tool and policy, so it should be used only as an order-of-magnitude sanity check when sizing a pilot, for example to see that tens of audio minutes may require tens of person-hours. It should not be

quoted as an expected cost. The pilot should measure its own annotation speed, and a payment rate should be set only after that measurement and after checking prevailing local rates. I do not propose a payment amount.

Sustainability questions remain open. Who funds contributors, who reviews, who adjudicates disputes, who holds the rights agreements, and who maintains the tool. A realistic first step is a small pilot funded by the author or a small grant, with all contributors fairly paid for their time, and with no commitments beyond the pilot.

9. Evaluation plan

This section answers RQ3. It is a protocol. No evaluation has been run, and no scores, dataset sizes, annotator counts, costs, or findings are reported in this paper. The thresholds and analyses below should be fixed in writing before any pilot annotation begins, and any later departure should be reported.

9.1 Questions the pilot should answer

1. Can contributors who have not been trained live complete the task from written guidelines and the interface, and where do they get stuck?
2. How closely do independently produced timestamps agree with each other and with an adjudicated reference?
3. How much annotation and review effort is needed per accepted audio minute, and what does that imply for cost?
4. Which acoustic and musical conditions produce the largest errors and disagreements?
5. Do automatic alignment suggestions reduce human effort for Amharic singing, or only add correction work?

9.2 Reference and sample

The reference annotation for each excerpt is an adjudicated one: two independent annotations are compared, differences above a pre-declared threshold are examined by an adjudicator who was not one of the two annotators, and the outcome is recorded. This reference is itself human judgement, so it is not ground truth, and its reliability is reported through the agreement between the independent annotations.

The pilot sample should be small, purposive, and stratified by conditions expected to matter: tempo, vocal style, density of accompaniment, presence of sustained notes, repetition, and code-switching. It is a feasibility sample, not a representative sample of Ethiopian music. As a planning suggestion only, a minimum useful design would include several songs per stratum, at least three annotators with differing experience, and double annotation of every excerpt. The actual size should be set after rights and annotator availability are confirmed, and not fixed in advance to a number that cannot be met. Because words within an excerpt and excerpts within a song are not independent, all confidence intervals should be computed by resampling excerpts or songs, not words, and strata should be reported descriptively unless the sample supports inference.

9.3 Metrics

Let a word w have reference boundaries (s_w, e_w) and compared boundaries (s'_w, e'_w), all in seconds. Table 6 defines the measures. Absolute errors are computed per boundary, onset and offset separately. Because the token list is frozen, words in the two annotations are paired one-to-one by index, with no alignment step.

Table 6. Proposed measures. No values are reported.

Measure	Definition	Purpose
Onset and offset error	|s'_w - s_w| and |e'_w - e_w| for each word.	Basic timing accuracy.
Mean absolute error (MAE)	Mean over all onset and offset errors, as in the average-absolute-error ranking measure of MIREX [1].	Comparability with the alignment literature.
Median and 90th percentile error	Computed separately for onsets and offsets.	Robust to outliers. Shows tails.

Measure	Definition	Purpose
Within-tolerance rate	Share of boundaries with error <= tau, for tau = 0.1 s, 0.3 s, 0.5 s. The 0.3 s value matches the MIREX tolerance [1].	Practical usability. Thresholds are fixed before analysis.
Correct-duration rate	Share of reference audio duration during which the compared word interval overlaps the reference word interval, following the MIREX percentage-of-correct-segments idea [1].	Interval-level agreement.
Inter-annotator agreement	The same error and tolerance measures computed between two independent annotators. Categorical flags (for example uncertain_time, melisma, language tag) use Cohen's kappa.	Reliability of the reference.
Correction rate	Share of words whose boundary changes by more than tau between first submission and adjudicated reference.	Review burden.
Annotation and review time per audio minute	Logged task time divided by excerpt duration. Reviewer time recorded separately.	Effort. Input to cost.
Acceptance and revision rates	Share of submissions accepted at first review and share returned for revision.	Process quality.
Cost per accepted minute	Formula of Section 8.4, using measured hours and an explicit rate.	Affordability.

Where transcript correctness is also evaluated, word error rate should be reported separately from timing error, since a correct timing of a wrong word is not a correct annotation. For automatic aligners, error against the adjudicated reference and the human time to correct the output are both reported, because an aligner that is slightly more accurate than another but harder to correct may not help.

9.4 Analyses and threats

The planned analyses are descriptive: distributions of error by boundary type, by stratum, and by annotator experience; where the largest errors occur (long vowels, repeated words, overlapping voices, masked consonants); and how guideline revisions change agreement across rounds. Several threats should be reported with the results. The annotators may be a small, self-selected group. The selected songs may be easier or harder than average. Annotators will learn the task and the guidelines will change, so early and late excerpts are not interchangeable. If the adjudicator and annotators work in the same small team, their judgements are not independent. And automatic aligners may have been trained on audio that overlaps with the test material, which can only be checked for models whose training data is documented.

A pilot of this size cannot support claims about all Ethiopian music, about Amharic singing in general, or about how an annotation market would behave at scale. If the pilot is not completed before the paper is circulated, this section should remain labelled as a planned protocol, and no hypothetical numbers should be presented in its place.

10. Rights, consent, and ethics

10.1 Recording rights and lyric rights

A song usually involves at least two layers of rights. The sound recording has its own rights holders (typically a performer, producer, or label), and the lyrics, as a literary and musical work, have their own (typically the author and any publisher or representative). Annotation touches both, because the timing file refers to both and a lyric transcript reproduces the text. That a recording can be streamed, or has been posted publicly, does not mean that it may be copied, stored in a dataset, redistributed, published with its lyrics, or used to train a model. Each of these uses should be treated as a separate permission to obtain and record.

10.2 Ethiopian law

The relevant national instrument appears to be the Copyright and Neighboring Rights Protection Proclamation No. 410/2004, which entered into force in 2004 and which WIPO Lex lists as amended by Proclamation No. 872/2014 and implemented by Council of Ministers Regulation No. 305/2014 [22]. I verified that these instruments exist and are listed, and I did not interpret their provisions. This paper therefore draws no legal conclusion about Ethiopian law. Questions that need local legal advice before any collection or release include the following.

- How rights in sound recordings, performances, and lyrics are allocated, how long they last, and who may license them.
- Whether any exception covers non-commercial research, text and data mining, or archiving, and whether it extends to a distributed dataset.
- Whether time-aligned annotations of a protected work are themselves derivative works or independent works, and who owns them.
- Whether performers have moral or performance rights that matter for releasing audio or training models.
- Which collecting societies or other bodies, if any, need to be consulted, and what contractual form a research licence should take.
- What applies when contributors, hosts, or users are in other jurisdictions, including those of the platforms the recordings came from.

10.3 Proposed collection and release policy

- Collect only from sources A to D of Table 5, with the permission basis and permitted uses recorded per recording and, separately, per lyric.
- Record provenance from audio source through transcript, annotation, review, and release.
- Release only what rights permit, in the narrowest form that serves research: annotations alone, annotations with controlled access to audio, or open release where the licence allows. A metadata-and-annotations release has narrower utility, and the dataset card must say so.
- Publish annotations separately from audio only after legal advice confirms that this is permitted. The annotation-only model of DALI [8] is a precedent for the technical form, not a legal clearance.
- Provide a takedown and correction route with a named contact and a stated response time, and record removals in the change log.
- Make no claim to grant an open licence over material the project does not control.

10.4 Contributor consent, privacy, and fair treatment

Contributors are told what they will do, how their work will be used and released, whether and how they will be paid, what data about them is kept, and how to contest a decision, before they start. Public releases use pseudonymous identifiers unless attribution is requested. Payment and identity data are stored separately and with restricted access. Rates are stated in advance, and rejected work receives a reason and a route to correction, in line with the concerns about crowd work raised by Fort et al. [21]. Local payment and tax obligations should be checked if the project grows.

10.5 Cultural context

Ethiopian music includes linguistic, regional, historical, secular, and religious traditions that should not be collapsed into one label. Religious material, including liturgical chant, calls for consultation with the relevant communities and should not be treated as interchangeable with popular song. Categories should rest on self-description or documented sources, not on guesswork, and contributors should be able to flag material that needs special handling.

10.6 Documentation

The eventual release should include a datasheet [19] and a data statement [20]. These ask for the motivation, composition, collection process, annotator demographics where appropriate and consented, language variety, intended uses, known limitations, licensing, and maintenance plan. The schema of Section 5 already collects most of what these require.

11. Limitations

This is a design paper, and its limitations are of three kinds.

Evidence. There is no pilot, no measured agreement, no annotation-cost data, and no evidence of how many contributors would participate or how reliable they would be. All annotation conventions are untested and likely to change. The effort benchmark of Section 8.4 is drawn from a different repertoire and task.

Review. The literature review is targeted, not systematic. It relied on web search and on reading source pages in October 2026, did not cover every relevant database or repository, and did not search Amharic-language sources. The conclusion that no contemporary Amharic word-timed song resource is public is therefore provisional. The availability and licence of the Zema Dataset, the language coverage of DALI, the coverage of Amharic in multilingual aligners, and the availability of Amharic resources for forced alignment were not established. Several claims about Amharic orthography, in particular the choice of the orthographic word as the unit, need confirmation by a fluent reviewer.

Design. Rights clearance may limit what can be released and may make the resource less useful than a fully open one. Automatic alignment may fail on Amharic singing, code-switching, dialectal pronunciation, or dense accompaniment, which would make the work mostly manual and therefore costly. Crowd annotation introduces its own errors. The prototype has no persistence, accounts, review queue, or logging of annotation time, and has not been tested with real users. It must not be used to collect real contributions at scale.

12. Future work

The next step is a rights-cleared pilot with a small group of fluent Amharic speakers, following the protocol of Section 9 and Appendix C. In parallel: contact the Zema authors about availability and conventions; check Amharic coverage in multilingual aligners; obtain legal advice on the questions of Section 10.2; add task-time logging, local save, and a reviewer view to the prototype; and have a fluent reviewer examine the tokenization and normalisation rules. After a pilot, the guidelines and schema can be revised and a small release prepared with a datasheet. Only after the workflow is validated should further steps be considered: other Ethiopian languages with language-specific conventions, backing-vocal layers, phoneme-level annotation, or musical layers such as melody and beat, each only when a clear use justifies its cost.

Potential users should be consulted before collection grows: speech and language technology teams, music-information-retrieval researchers, lyric and karaoke developers, music educators, and Ethiopian language-technology groups. Their needs determine which fields matter, what licence is acceptable, and what accuracy is useful. Applications such as synchronised lyrics, karaoke, lyric search, and music education are possible downstream uses, and none has been demonstrated by this work.

13. Conclusion

This paper set out a design for word-level time-aligned lyrics for contemporary Amharic songs. It places the idea beside existing work: word-level alignment is an established task with public benchmarks in other languages, and Ethiopian vocal music already has a word-timed resource in liturgical chant. Within that landscape, a narrower claim is supportable. In the sources reviewed, a public, documented collection of contemporary Amharic songs with word timings was not found, and the conventions, reliability, effort, and rights handling for building one have not been worked out in the literature reviewed.

The proposal consists of a draft schema, annotation conventions for difficult cases, a ten-step rights-aware workflow, a minimal architecture, a proof-of-concept annotation interface, and an evaluation protocol with pre-declared metrics. Of these, only the schema draft and the prototype exist as artefacts, and neither has been tested

with annotators. The most valuable next contribution is a small, rights-cleared pilot, whose results would turn this design into evidence.

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Appendix A. Release checklist

- Rights evidence recorded for every recording and, separately, every lyric, with permitted uses stated.
- Legal advice obtained on the questions in Section 10.2 before any public release.
- Audio duration and checksum validated. Every timestamp passes the structural checks.
- Lyric text and tokenization checked by a fluent speaker. Tokens frozen before timing.
- Independent second annotation and adjudication completed for every released excerpt, with agreement reported.
- Schema, licence, version number, change log, citation text, datasheet, and data statement published.
- Contributor personal data excluded. Pseudonymous identifiers only, unless attribution was requested.
- Takedown and correction contact published, with a stated response time.
- All example data marked as illustrative, and no illustrative data included in a release.

Appendix B. Draft annotation guidelines (version 0.1)

These guidelines are written in the form an annotator would see. They are untested and will change after the pilot.

B.1 Your task

You will hear a short excerpt and see its lyric, already split into words. For every word you set a start time and an end time. You do not change the words. If you think a word is wrong or missing, mark it as uncertain and add a note.

B.2 What start and end mean

- **Start:** the earliest moment at which you can clearly hear the word begin. Choose the earliest clear evidence, not the moment you first guess it.
- **End:** the last moment at which you clearly hear the word, including a sustained vowel, before the next word begins or the voice stops.
- If two boundaries are both defensible, choose according to the two rules above and mark the word uncertain_time.

B.3 Workflow for each excerpt

1. Listen to the whole excerpt once without editing, following the lyric.
2. Time the words in order. Use the playhead buttons to set boundaries, then replay the word to check it.
3. Use the numeric fields only for fine adjustments of a few milliseconds.
4. Mark any word you are unsure of. Do not guess.
5. Check the validation messages. Fix structural errors. Export or submit.

B.4 Special cases

See Table 4 for the rules on held notes, repeated words, overlapping voices, code-switching, instrumental breaks, elision, spoken words, and ambiguous boundaries. Two reminders: every sung repetition is its own word, and instrumental sections get no words.

B.5 What not to do

- Do not edit, re-spell, or normalise words.
- Do not copy another annotator's times or ask for them.
- Do not work on recordings you are not given.
- Do not rush. Accepted work is judged on accuracy, not speed.

Appendix C. Pilot protocol and evaluation checklist

C.1 Before the pilot

1. Fix and date in writing: tolerances, metrics, analysis plan, sampling strata, and what counts as a disagreement needing adjudication.
2. Obtain rights evidence for every excerpt. Do not start annotation of any excerpt without it.
3. Prepare verified lyrics and freeze tokens. Select an adjudicator who will not annotate the same excerpts.
4. Recruit annotators with a range of experience. State payment, review time, and appeal route in advance.
5. Add task-time logging to the interface and test the whole workflow on one excerpt.

C.2 During the pilot

1. Have at least two annotators time every excerpt independently, without seeing each other's work.
2. Record annotation time, review time, number of boundary edits, and revision and acceptance outcomes.
3. For a subset or all excerpts, also run the automatic aligners under comparison and record model version and settings, then measure correction time.
4. Adjudicate disagreements above the declared threshold. Record the rule used and revise guidelines in numbered versions.

C.3 After the pilot

1. Compute the measures of Table 6 per excerpt, with confidence intervals obtained by resampling excerpts or songs.
2. Report by stratum and by annotator experience descriptively. Do not draw general conclusions from a few songs.
3. Report cost per accepted minute from measured hours and the rate actually paid.
4. Report all departures from the written plan, all failures, and all limitations.

Appendix D. Source verification and search log

This appendix records what was checked for this version, how, and what remains. "Page read" means the source page or record was retrieved and read in October 2026. It does not mean the claims in the source were independently confirmed.

Table D1. Verification status of references.

Ref.	Source	Checked	Still to do
1	MIREX 2026 task page	Page read. Task, datasets, metrics, edit date.	Re-read if cited after the page changes.
2	Mesaros and Virtanen, 2010	Springer record read: title, authors, year, article number, abstract.	None for the claims used.
3	Schulze-Forster et al., 2021	DOI resolves to the IEEE Xplore record. Title and authors from search results. Volume and pages removed from the reference because they could not be confirmed.	Add volume and pages from IEEE Xplore if you can open it.
4	Stoller et al., 2019	arXiv record read. Title, authors, date, abstract. Cited as an arXiv preprint because the venue could not be confirmed.	Add ICASSP 2019 details if you confirm them.
5	Durand et al., 2023	arXiv record read, including ICASSP DOI. Claims reported as the authors' own.	None for the claims used.

Ref.	Source	Checked	Still to do
6	Cífka et al., 2024	arXiv record read. ISMIR 2024 programme page read and shows the same title and authors.	Paper type (full or late-breaking) not stated on that page.
7	JamendoLyrics dataset card	Page read: 79 songs, 4 languages, per-song licences, creators.	Check licence terms of any song actually used.
8	DALI, TISMIR 2020	Journal page read: authors, volume, pages, DOI, size, contents.	Check whether Amharic is present.
9	Gupta et al., ISMIR 2018	Zenodo record read. It is a conference paper, not a dataset.	Not used for any dataset claim.
10, 11	Zema paper and GitHub repository	arXiv full text read for size, annotation process, language, availability. Repository page read: contents and absence of a licence file.	Contact authors. Confirm what is downloadable and under what terms. Peer-review status of the paper.
12	Dataset.ET blog post	Page read: date, size, licence, host, speech type.	Cite a peer-reviewed or archival description if one appears.
13	Abate et al., LREC 2020	ACL Anthology page read. Authors, pages, WER.	Confirm from the PDF whether 22 h is per language.
14	Gete et al., 2025	arXiv record read.	Check whether a published version exists.
15	Belay et al., 2021	PDF read for content. ICT4DA 2021 comes from two bibliographic listings; the PDF itself does not state it.	Confirm venue and pages from the proceedings.
16	Pratap et al., 2024	JMLR page and arXiv text read for the 1,130-language alignment statement.	Check whether Amharic is included and how to use the aligner.
17	McAuliffe et al., 2017	ISCA archive page read.	Check Amharic resources if used.
18	lyrics-aligner repository	README read: ARPAbet input, limitations, licence.	None.
19, 20	Datasheets for datasets. Data statements	Gebru: Crossref record read (CACM 64(12), 86-92). Bender and Friedman: ACL Anthology record read.	None.
21	Fort et al.	ACL Anthology record and PDF read. Wages below US$2 per hour, no reliable gain in quality from higher pay, and calls for fair pay and ethics review match what the paper says.	None for the claims used. A Mason and Watts reference was removed because its claim could not be re-checked.
22	WIPO Lex record	Page read: law title, year, amendments, regulation.	Obtain and read the full texts with a lawyer.

Searches performed (October 2026, web search and page reading). Zema dataset availability; Amharic and Ethiopian song or lyric alignment datasets; Ethiopian copyright law records; multilingual speech and forced alignment; Jam-ALT and JamendoLyrics; Amharic speech corpora and ASR; Amharic homophone normalisation; crowdsourcing incentives and ethics; datasheets and data statements; forced-aligner papers; ISMIR 2026 submission calls. The search for Amharic or Ethiopian song-alignment datasets was repeated on 14 October 2026 and again found none (results were English or other-language alignment work and Amharic read speech).

Not searched or not exhausted. A systematic search of ISMIR, ICASSP, Interspeech, LREC, and ACL Anthology proceedings; Google Scholar and Scopus; Hugging Face and Zenodo dataset listings for Amharic or Ethiopian music; Ethiopian university repositories; Amharic-language sources; and the full text of DALI,

JamendoLyrics, and the multilingual singing datasets for Amharic content. Any of these could change the gap statement of Section 3.1.